

Calcium, Parathyroid Hormone, and Blood Pressure

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This review critically analyzes the available information on the relationship among calcium, parathyroid hormone, and blood pressure regulation. Both acute and chronic hypercalcemia increase blood pressure primarily via direct effects on vascular smooth muscle contractility. The evidence for indirect effects through activation of hormonal pressor systems is inconclusive.

In apparent contrast with the notion that hypercalcemia can cause hypertension, more recently it has been proposed that calcium deficiency may be important in the genesis of hypertension both in humans and in spontaneously hypertensive ani-

mals. However, the evidence supporting this notion is still conflicting.

Parathyroid hormone exerts complex actions on the cardiovascular system. On one hand, if injected in pharmacological doses, it is a vasodilator and antagonizes the pressor action of norepinephrine and angiotensin II; on the other hand, parathyroid hormone can potentiate the pressor effect of hypercalcemia. *Am J Hypertens* 1989;2:34S-44S

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The role of calcium in blood pressure regulation is quite complex; it is currently the subject of animated disputes in the literature.¹⁻³ A large body of evidence indicates that both acute and chronic hypercalcemia increase blood pressure. More recently, however, McCarron¹ has suggested that decreased dietary calcium intake and hypocalcemia, through a chain of events largely unexplained, can result in hypertension. Whereas the evidence linking hypercalcemia to hypertension is convincing and undisputed, the evidence linking dietary calcium deficiency with hypertension in man is equivocal and inconclusive. The purpose of this review is to critically analyze the available information on the relationship among calcium, parathyroid hormone, and hypertension.

HYPERCALCEMIA AND BLOOD PRESSURE

Several lines of evidence indicate that both acute and chronic hypercalcemia can cause hypertension and that correction of the hypercalcemia is often followed by normalization of blood pressure. Acute administration of calcium increases blood pressure both in normal human subjects,⁴ as well as in rats,^{5,6} and a close relationship exists between the increments in the concentration of serum calcium and the rise in blood pressure.^{5,6} Acute hypocalcemia, induced by the administration of edetic acid (EDTA), can also cause significant perturbation in blood pressure regulation, particularly during upright posture; a significant correlation has been shown between the orthostatic fall in blood pressure and the decrease in serum calcium.⁷

Chronic hypercalcemia has also been associated with increased prevalence of hypertension in a variety of clinical conditions. Those include primary hyperparathyroidism,⁸⁻¹⁰ secondary hyperparathyroidism with hypercalcemia, (which may occur in chronic renal failure or after renal transplantation,^{11,12} vitamin D in-

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toxication,¹³ prolonged immobilization,¹⁴ and hypercalcemia associated with malignancy.

The pathophysiological mechanisms linking hypercalcemia to hypertension are complex. Berl et al⁶ observed that acute hypercalcemia in conscious rats caused a significant increase in peripheral vascular resistance without changes in cardiac index. Pretreatment with calcium channel-blockers inhibited the pressor effect of hypercalcemia. Marone et al¹⁵ observed similar hemodynamic changes in normal or hypertensive human subjects. No consistent changes in cardiac output have been observed during calcium infusion.^{15,16} Studies on the effect of calcium on perfusion pressure in the forelimb, kidney, and heart in the dog, have shown that this cation causes arteriolar constriction and increased myocardial contractility.^{17,18}

The vasoconstrictor action of calcium can be both direct and indirect. Ca^{2+} plays a central role in excitation contraction coupling of vascular smooth muscle cells. Regardless of the mechanism causing cell excitation, whether it is neural, humoral, or myogenic, it results in a rise in cytosolic Ca^{2+} concentration. The increased intracellular Ca^{2+} activates the calmodulin-dependent myosin light chain kinase, leading to formation of cross bridges between the actin and myosin filaments and to cellular contraction.

An increase in extracellular calcium during acute or chronic hypercalcemia could enhance transcellular flux of calcium, ultimately resulting in vasoconstriction. The phasic contraction in isolated femoral artery strips from spontaneously hypertensive rats (SHR), deoxycorticosterone acetate (DOCA) hypertensive rats, and 2K-1C renal hypertensive rats, was markedly increased when the calcium concentration was increased from 1.6 mmol to 3.2 mmol.¹⁹

Increased extracellular concentration of Ca^{2+} may also increase the vascular responsiveness to norepinephrine (NE) or angiotensin II,^{17,19} or to potassium,²⁰ in isolated vascular preparations. Renal vascular sensitivity to NE decreases greatly either when the extracellular calcium concentration is reduced or in the presence of a calcium antagonist.²¹ Perfusion of the rat hindquarters with a Ca^{2+} free solution abolished the vasoconstrictor response to NE.²¹ Furthermore, calcium channel-blockers such as nifedipine, verapamil, and diltiazem antagonized the pressor effect of NE, angiotensin II, and vasopressin in rat.²² Ca^{2+} is also important in the binding of NE to α -adrenergic receptors in the rabbit aorta.²³

Conflicting data are available on the effect of hypercalcemia on vascular response to NE in human subjects. On one hand, Marone et al¹⁵ have shown that acute hypercalcemia, induced by intravenous administration of calcium, did not alter the pressor response to NE or angiotensin II in normal subjects. Vlachakis et al,²⁴ on the other hand, have shown that chronic hypercalcemic

patients display enhanced blood pressure response to NE.

Hypercalcemia can also raise blood pressure through its effects on NE release from the sympathetic end terminals and from the adrenal glands.²⁵⁻³¹ According to the "calcium hypothesis",^{25,29} depolarization of the sympathetic end terminals increases Ca^{2+} influx, which stimulates NE exocytosis. It has been postulated that the endoskeleton of the sympathetic nerve terminals contains actomyosin-like neurofilaments capable of contraction when the intracellular concentration of Ca^{2+} increases. During the contraction of these neurofilaments, vesicles containing neurotransmitters migrate towards the membrane. The release of NE from the perfused cat spleen or from isolated spleen slices in response to nerve stimulation was abolished if the perfusate was calcium free,³² and it was potentiated when the concentration of calcium was modestly increased.^{33,34} However, at concentrations over 7 mEq/L, calcium increased the stability of neuronal membrane and inhibited NE release.³⁵ On the contrary, the basal efflux of NE from rat mesenteric arteries is stimulated by reduced concentration of Ca^{2+} .^{33,36}

Very scanty and inconclusive data on the effect of Ca^{2+} on catecholamine release is available in human subjects. Marone et al¹⁵ have shown that IV administration of calcium increases plasma NE, but this occurred at a time when volume depletion was also evident. Vlachakis et al²⁴ have shown increased plasma levels of NE in patients with hypercalcemia secondary to primary hyperparathyroidism, but they did not measure blood volume in their patients. Furthermore, these studies can not rule out a possible effect of PTH on plasma catecholamines.^{37,38} In conclusion, the evidence linking calcium to NE release both in humans and experimental animals is conflicting and inconclusive.

The mechanisms involved in the increase in blood pressure during acute hypercalcemia are largely independent of the renin-angiotensin system. An inverse relationship exists between intracellular Ca^{2+} and the rate of renin release. A decrease in intracellular Ca^{2+} , caused by decreased influx or increased efflux, stimulates renin secretion, whereas a rise in intracellular calcium inhibits renin secretion.^{39,40} On the contrary, aldosterone secretion in vitro requires calcium.⁴¹⁻⁴³ Despite these in vitro observations, studies in humans have shown no change in plasma renin or aldosterone levels during acute hypocalcemia,⁷ or hypercalcemia.¹⁵

CALCIUM DEFICIENCY IN HYPERTENSION

In apparent contrast with the notion that acute or chronic hypercalcemia can cause hypertension, more recently it has been proposed that calcium deficiency may be important in the genesis of hypertension both in humans and in SHR.¹

This hypothesis stems from the observation that a wide range of derangements in calcium metabolism can be observed in patients with essential hypertension or in SHR (Table 1). According to this hypothesis, a decreased dietary intake of calcium and/or hypercalciuria leads to decreased serum concentration of Ca^{2+} ; this derangement potentiates Ca^{2+} fluxes into smooth muscle cells, thus, causing vasoconstriction. Let us review the evidence supporting this hypothesis.

The Evidence for Hypercalciuria There is little dispute in the literature about this abnormality in calcium metabolism in hypertension. Hypercalciuria has been shown both in patients with essential hypertension,^{44–46} as well as in SHR.^{47–49} A great deal of controversies exist, however, over the causes of this abnormality. Theoretically, hypercalciuria could be the consequence of increased dietary intake, increased intestinal absorption, or of a renal tubular leak of calcium; it could also be related to increased natriuresis caused by higher dietary sodium intake.

There is no evidence of increased dietary intake of calcium in hypertensive subjects. To the contrary, several epidemiologic studies have suggested that hypertensive patients may, on the average, ingest less calcium in their diet.¹ No study has yet been published showing abnormalities of intestinal calcium absorption in hypertensive patients, although the data on intestinal calcium absorption in SHR are conflicting.^{50–52}

McCarron¹ has also suggested that the hypercalciuria in SHR is dependent on a renal tubular calcium leak. Lau et al,⁴⁹ on the other hand, using clearance techniques in rats with intact parathyroid glands or parathyroidectomies, were not able to detect a calcium leak after an overnight fast.

One final cause of hypercalciuria in essential hypertension could be increased dietary sodium intake and natriuresis, because the urinary excretion of calcium is closely correlated with the urinary excretion of sodium.^{53–56} McCarron,¹ however, has shown that hypercalciuria in essential hypertension is present at any level of urinary sodium excretion. Despite the strong and convincing appeal of this argument, there are equally valid counterarguments. Under steady-state conditions, hypercalciuria can also be found in two

completely different models of experimental hypertension, the DOCA-salt,⁵⁷ and the Dahl's salt-sensitive rats;⁵⁸ both these models are characterized by expansion of body fluid volume. In the DOCA-salt model, urinary sodium excretion falls on the first day of DOCA treatment and then returns to baseline value; in contrast, urinary calcium excretion increases only after two to three days probably as a consequence of volume expansion. This expansion would decrease proximal tubular reabsorption of calcium and sodium and increase distal delivery; mineralocorticoids would promote distal tubular reabsorption of sodium but not of calcium, thus resulting in hypercalciuria. One can, therefore, envision a similar chain of events in human hypertension. According to Blaustein,⁵⁹ hypertension is related to a genetic inability of the kidneys to excrete a sodium load. In the early phase of the development of hypertension, this defect may lead to volume expansion and to release of ouabain-like substances which, by inhibiting the Na,K-ATPase pump, would offset the genetic renal defect and result in normalization of urinary sodium excretion and of sodium balance. The inhibition of Na,K-ATPase at the level of vascular smooth muscle cells would, however, result in increased intracellular sodium and in reduced calcium extrusion, ultimately leading to increased vascular tone. The slight volume expansion and hypertension could sustain hypercalciuria despite normal urinary sodium excretion with a mechanism similar to the one observed in DOCA-salt treated dogs.⁵⁷

A further argument against the hypothesis that hypercalciuria exerts a causative role in hypertension, is derived from the clinical observation that patients with primary hypercalciuria do not appear to be at greater risk of developing hypertension.⁶⁰

Intestinal Calcium Absorption In Hypertension

McCarron¹ has suggested that patients with essential hypertension, as well as rats with genetic hypertension, display a defect in intestinal absorption of calcium and this derangement, in concomitance with decreased dietary calcium intake, leads to negative calcium balance and to hypocalcemia. Schedl et al⁵⁰ and Lucas et al⁵¹ have indeed demonstrated a decrease in maximal transport rate and net total transport of calcium in SHR. Other investigators,⁵² however, have observed increased intestinal absorption of calcium, positive calcium balance, and increased serum levels of $1.25(\text{OH})_2\text{D}_3$ levels in prehypertensive three-week-old SHR. Other found no abnormality of Ca^{2+} uptake by everted gut sacs in vitro.⁶¹ The reason for these discrepancies are not readily apparent, but it may be related to different experimental protocols. It has also been suggested that an alteration in vitamin D metabolism may be responsible for the abnormality in intestinal calcium transport in these animals.⁵⁰ No data on intestinal calcium absorption are available in human subjects. Abnormalities in

TABLE 1. DERANGEMENTS IN CALCIUM METABOLISM IN HYPERTENSION

1. Hypercalciuria
2. Decreased intestinal absorption of calcium
3. Low calcium intake in hypertensive patients
4. Reduced serum ionized calcium
5. Increased cytosolic Ca^{2+}
6. Increased serum levels of parathyroid hormone (PTH)
7. Abnormal levels of Vitamin D_3 and calcitonin
8. Hypotensive effects of oral calcium loading
9. Abnormalities in cell membrane transport of calcium

Vitamin D metabolism have been reported in patients with low-renin essential hypertension who also manifest hypocalcemia.⁶² If the hypothesis of McCarron was correct, ie, of decreased dietary intake and intestinal absorption of calcium, and of hypercalciuria, hypertensive patients and SHR should display a significant negative calcium balance. There is no evidence of increased osteopenia in patients with essential hypertension, nor in SHR; on the contrary, as we have previously stated, SHR at least in one study, manifested increased skeletal content of calcium.⁴⁹

Reduced Calcium Intake in Hypertension McCarron¹ claimed that there is an inverse relationship between calcium intake and the prevalence of hypertension in a given population; the lower the dietary calcium intake, the greater the probability of developing hypertension. This claim is based on analysis of the U.S. National Health and Nutrition Examination Survey I (NHANES I).⁶³

The analysis performed by McCarron et al,⁶³ as well as their interpretation, has met with serious criticisms. This analysis was based on 24-hour dietary recalls, a method that has major shortcomings, because it is very difficult to validate and also dietary intakes may change on a daily basis.⁶⁴ They also found an inverse relationship between body mass index and dietary calcium intake that could indicate that overweight patients may consume less dairy products in order to lose weight or, alternatively, it might suggest that overweight patients may be less willing to fully reveal their dietary habits. It is, in fact, surprising that in the analysis of McCarron et al⁶³ the average intake of sodium was of only 90 mEq/d, far less than the average sodium intake in the American population assessed by determination of urine excretion.^{65,66} The analysis of McCarron et al⁶³ has also been criticized, because it excluded subjects with a history of hypertension. Furthermore, using the same data from the NHANES I survey, other investigators have reached different conclusions. Feinleib et al⁶⁷ found no difference in dietary calcium intake between normotensive and hypertensive subjects when age and body weight were taken into account. Harlan et al,⁶⁸ on the other hand, found an inverse correlation between dietary calcium intake and diastolic blood pressure in women but a direct correlation with systolic blood pressure in men. At least two other epidemiologic studies^{69,70} have failed to show any direct relationship between dietary calcium intake and prevalence of hypertension. Finally, the prevalence of hypertension appears to be lower in unacculturated societies despite their low calcium intake.⁷¹ In conclusion, evidence for a relationship between dietary calcium intake and prevalence of hypertension is weak and inconclusive.

Several laboratories have observed higher blood pressure in normotensive and hypertensive rats during

long-term calcium deprivation.^{72–74} However, these studies did not take into account the fact that rats with very low calcium intake also develop magnesium depletion, hyperparathyroidism, and growth retardation—variables that are known to affect blood pressure control.^{75,76} Mild calcium restriction within the physiological range did not result in increased blood pressure in SHR.⁷⁷

The Evidence For Hypocalcemia McCarron⁷⁸ was the first to report decreased serum-ionized calcium but not total calcium in hypertensive subjects. Resnick et al,⁷⁹ on the other hand, observed decreased ionized calcium only in patients with low renin, or in salt-sensitive patients,⁸⁰ but not in those with normal or high renin; others, however, could not confirm difference in serum calcium in patients with low renin.⁸¹ Others have shown normal serum-ionized,^{82,83} and total calcium,⁸² or lower levels of total calcium,⁸⁴ or increased plasma calcium in essential hypertension.^{85–89} Finally, Harlan et al,⁶⁸ and Kesteloot et al⁸⁶ have shown a direct relationship between serum calcium and blood pressure among patients with essential hypertension in spite of an inverse relationship between dietary calcium intake and blood pressure. Salivary-ionized calcium has been shown to be decreased in subjects with essential hypertension.⁸⁹ Data on intracellular content on calcium in platelets or leukocytes are also inconsistent. Erne et al⁸⁴ found a direct correlation between content of calcium in platelets and blood pressure, whereas others could not confirm these findings.^{90,91} Cytosolic free calcium in neutrophils was also not different in hypertensive subjects.^{92,93}

The reason for these discrepancies in calcium levels are difficult to interpret, but they could be related to the fact that multiple variables can affect serum calcium such as serum albumin, the volume-sodium status, blood pH, alcohol intake, muscular exercise, respiratory alkalosis, fasting *v* feeding, hemoconcentration, diuretic therapy, age, and sex of the patients.^{94–97} It is obvious that balance studies under rigid metabolic control and experimental conditions are needed to ascertain whether these hypothetical abnormalities in serum calcium are present in patients with essential hypertension.

The evidence for reduced total calcium or serum-ionized calcium is more consistent in genetically hypertensive rats.^{47–99} Whether this abnormality has any pathogenetic meaning or is simply the result of hypertension remains to be determined. It is of significance that hypocalcemia is also observed in other forms of experimentally induced hypertension, such as in the DOCA-salt and in the 5% nephrectomy.⁹⁸ These observations would rather suggest that hypocalcemia might be a secondary rather than a primary phenomenon.

Lau et al,⁴⁹ on the other hand, considering that rats are nocturnal animals, measured serum-ionized calcium in

SHR at midnight after their calcium intake with meals and found increased levels compared with WKY.

The Proposed Link Between Hypocalcemia And Hypertension McCarron¹ has suggested that hypocalcemia may inhibit Ca-ATPase activity that, in turn, would lead to increased intracellular Ca^{2+} and to smooth muscle cell contraction. An opposite effect would raise levels of serum Ca^{2+} .

These authors¹ and others^{100–103} have indeed shown decreased Ca-ATPase pump activity in vascular smooth muscle cells or in RBC, of SHR and humans with essential hypertension. Increased levels of serum Ca^{2+} would have the opposite effect. Similar observations have been made in DOCA-salt rats.¹⁰⁴ This may lead to increased permeability to calcium and to a rise in cytosolic free calcium.^{105–108} Membrane uptake of calcium depends on calcium itself, and on calmodulin,^{109,110} two factors that, in turn, can activate Ca-ATPase. Calcium supplementations would revert these abnormalities and lead to stabilization of the vascular smooth muscle cells and to decreased blood pressure. To further support their contention, McCarron¹ cites the work of Bohr et al¹¹¹ and the work of Overbeck.¹¹²

Bohr et al¹¹¹ and Webb et al¹¹³ observed that an increase in the concentration of calcium in the medium, leads to a decreased response of helical strips of rat tail arteries to constrictive agonists. These investigators also have demonstrated that concentrations of calcium chloride in the medium between 2 and 4.1 mmol/L potentiate the vascular response to NE, whereas concentrations greater than 8 mmol/L decrease the contractile response to NE. These actions of high concentration of calcium were attributed to a “membrane-stabilizing” effect. Holloway et al¹¹⁴ have also shown that in isolated arterial segments from hypertensive rats, the concentration of calcium required to cause relaxation is higher than in arteries from normotensive animals.

Overbeck,¹¹² using an isolated hindlimb preparation from the New Zealand genetically hypertensive rat and its normotensive control, has shown progressive vasodilatation when the concentration of calcium in the perfusate was raised from 2 to 30 mEq/L. However, in other studies in dogs,¹¹⁵ and in humans,¹¹⁶ Overbeck found that calcium infusion caused vasoconstriction and not vasodilatation.

It is obvious that these observations from experiments in vitro can hardly be extrapolated to substantiate assumptions in human hypertension. The concentrations of calcium used to demonstrate a membrane-stabilizing effect in vitro, are extremely higher than the physiological concentrations of this element. Furthermore, the changes in serum Ca^{2+} observed in humans or experimental hypertension are much smaller than those necessary to demonstrate effects on vascular contractility in vitro.

Winqvist et al,¹⁷⁷ in separate experiments, have shown increased spontaneous contractile activity in vascular preparations from hypertensive animals. They found that the magnitude of spontaneous rhythmic contraction was greater when the concentration of calcium was increased from 1.6 mmol/L to 3.2 mmol/L. Several studies have also shown that, following activation by certain contractile agonists, the vascular smooth muscle cells of hypertensive arteries are more sensitive to calcium ions.^{118,119} Finally, the concept of a link between hypocalcemia and hypertension is in apparent contradiction with the observations that acute as well as chronic hypocalcemia may lower rather than increase blood pressure.⁷

Effect Of Calcium Supplementation On Blood Pressure If hypocalcemia is a causative factor in human or in congenital hypertension, calcium supplementation should normalize serum calcium and blood pressure. The chronic administration of 1.75% to 4.3% calcium in the diet attenuates the development of hypertension in SHR.^{47–120} This effect has been attributed by some to phosphate depletion caused by high dietary calcium intake.¹²¹ In fact, parenteral phosphate administration reversed the antihypertensive effect of calcium.¹²¹ However, phosphate added to the diet did not reverse the antihypertensive effect of dietary calcium.¹ Stern et al,⁶¹ on the other hand, observed that calcium supplementation corrected the abnormal serum Ca^{2+} and PTH but did not alter the progression and severity of hypertension in SHR. Even less conclusive are the data on the effect of calcium supplementation on blood pressure in human subjects.

Belizan et al^{122,123} have shown that a large amount of calcium (1 g/d) caused a significant decrease in diastolic blood pressure in normotensive subjects. McCarron and Morris,¹²⁴ however, have shown no effect of dietary calcium supplementation on blood pressure in normal subjects. However, these investigators observed a small but significant decrease in blood pressure in untreated patients with essential hypertension, after dietary supplementation with calcium (1 g/d) for eight weeks. The decrease in blood pressure was significant only during upright posture. Because no measurements of blood volume were obtained in these studies and increased dietary calcium may increase natriuresis,⁶¹ it is possible that the mild orthostatic changes in blood pressure caused by calcium supplementation may be related primarily to intravascular volume depletion rather than to effects on smooth muscle cell contraction.

Other investigations^{125–128} have been unable to confirm a fall in blood pressure with calcium supplementation. Resnick et al¹²⁹ observed a reduction only in low-renin or in salt-sensitive patients with essential hypertension. Grobbee and Hofman¹³⁰ showed a fall in

blood pressure during oral calcium intake only in patients with low serum Ca^{2+} and high PTH levels. It is conceivable that in these patients calcium supplementation may cause natriuresis with subsequent decrease in blood pressure. Unfortunately, no metabolic balance studies have been performed to address those issues.

It is obvious that more studies are needed before a recommendation can be made to all hypertensive patients to increase dietary calcium intake. This is particularly important, because any attempt to manipulate diets in order to increase the intake of calcium would entail an increment in dietary intake of fat, calories, and salt that could ultimately worsen cardiovascular morbidity and mortality.

Role Of Parathyroid Hormone In Blood Pressure Regulation Parathyroid hormone (PTH) exerts complex actions on the cardiovascular system. On one hand PTH, in pharmacological doses, has been shown to act as a vasodilator; on the other hand, this hormone can potentiate the pressor effect of hypercalcemia. Several studies have shown that PTH is a vasodilator. Both the purified bovine PTH (1-84 PTH) and its synthetic aminoterminal fragment (1-34 PTH) increase coronary, renal, hepatic, gastric, and pancreatic blood flow in the dog^{131,132} and decrease blood pressure in a variety of normotensive anesthetized vertebrates.^{133,134} Acute administration of 1-84 or 1-34 PTH in SHR, in DOCA-salt rats, and in rats with the two-kidney—one-cup model of hypertension reduced blood pressure in a dose-dependent fashion.^{135,136} Both 1-84 and 1-34 PTH also inhibit the pressor effect of NE and angiotensin II in normal rats, and this action of the hormone is totally abolished by pretreatment with indomethacin.¹³⁷ Infusion of 1-84 and 1-34 PTH produced a significant rise in urinary excretion of 6-keto-PGF₁. These data indicate that the vasodilatory effect of PTH is mediated by increased production of prostacyclin. Because the calcium ionophore A23187 enhances prostaglandin synthesis in platelets¹³⁸ and in the renal medulla,¹³⁹ the vasodilatory action of PTH may be due to increased production of prostacyclin stimulated by increased calcium flux in smooth muscle cells. Parathyroid hormone increases the intracellular uptake of calcium in a variety of tissues.^{140–142} However, it is not clear whether PTH also increases calcium flux into vascular smooth muscle cells. Yanc et al¹⁴³ and Pang et al¹⁴⁴ have shown that PTH may block the entry of calcium into smooth muscle cells. These authors have also shown that 1-34 PTH reduces the low-affinity lanthanum-resistant pool of calcium in the rat tail artery and in the mesenteric artery of the chicken.¹⁴⁴ Schleiffer et al,¹⁴⁵ on the other hand, have shown that PTH increased intracellular calcium uptake in rat aorta and cardiac cells.

Whether the vasodilatory action of PTH is of clinical importance is still a question. Increased levels of PTH

have been found in hypertensive patients as well as in SHR,¹ probably in response to reduced levels of Ca^{2+} . McCarron¹ has suggested that the increased levels of PTH may exert a protective role on the rise in blood pressure caused by the hypocalcemia. However, some studies have shown a significant change,⁵ or actually a decrease of blood pressure in normal animals following the removal of all parathyroid glands even when the serum calcium was maintained at normal levels with oral calcium supplementation. Furthermore, parathyroidectomy has been shown to attenuate the levels of hypertension in SHR and in DOCA-salt rats.^{146,147} Finally, hypertension is not more common in patients with hypoparathyroidism.⁶⁰

Increased levels of parathyroid hormone may be exerting an important role in the genesis of certain cardiovascular abnormalities observed in uremia. Patients and rats with chronic renal failure have secondary hyperparathyroidism, and they also manifest reduced responsiveness to the vasoconstrictor action of NE and angiotensin II.^{148,149} This abnormality can be abolished by administration of indomethacin¹⁵⁰ or by parathyroidectomy.¹⁴⁹ These data suggest that the state of secondary hyperparathyroidism may be responsible for the reduced pressor response to NE in uremia, via stimulation of prostacyclins.¹³⁷

As mentioned earlier, PTH, despite its vasodilatory action, under conditions of hypercalcemia appears to potentiate the vasoconstrictor response of calcium. Several observations substantiate this contention. First, hypertension is more common in patients with primary hyperparathyroidism than in the general population.¹⁰ Second, acute hypercalcemia raised blood pressure in normal rats. However, for a comparable rise in serum calcium, the increment in mean arterial pressure (MAP) in parathyroidectomized (PTX) rats was lower than in control animals.¹⁴⁹ When PTH was infused intravenously to the PTX rats, the blood pressure response to calcium infusion was normalized. Third, infusion of calcium chloride, to patients subjected to parathyroidectomy caused a lesser rise in blood pressure than in normal subjects.³⁸ Fourth, long-term administration of PTH can increase blood pressure in humans and in rats.^{151,152} Fifth, parathyroidectomy attenuates hypertension in SHR and DOCA-salt rats.^{146,147} Sixth, for a comparable rise in serum calcium, blood pressure increased much more in patients¹⁵¹ or in rats¹⁵² with chronic renal failure, probably as a result of the state of secondary hyperparathyroidism.

Because PTH has been shown to enhance calcium entry into many organs and tissues, it can be speculated that in the presence of PTH and hypercalcemia, more calcium enters the smooth muscle cells, thus causing more vasoconstriction. One must also speculate, that in the presence of hypercalcemia, the vasoconstrictor action of PTH overcomes its vasodilatory effect, which

appears to be also related to the ionophoric action of this hormone.

There is also some circumstantial evidence to suggest that PTH might affect blood pressure through some action on the sympathetic nervous system. Ionophores, which directly cause the entry of Ca^{2+} bypassing membrane potential changes, stimulate the exocytosis of NE.³⁷ Thus, PTH could also increase Ca^{2+} influx into the neuroplasm, leading to NE release. The evidence supporting this concept is indirect and inconclusive. Genari et al³⁸ have shown that calcium infusion causes a rise in plasma NE levels in normal subjects but not in parathyroidectomized patients; this suggests that a permissive role of PTH may be needed for the stimulation of NE release caused by acute hypercalcemia.

Vlachakis et al²⁴ have also demonstrated higher levels of plasma catecholamines in patients with primary hyperparathyroidism; it is not clear, however, from those studies whether the rise in plasma catecholamines is due to the hypercalcemia, changes in blood volume, or to the hyperparathyroidism. More studies are needed to determine if PTH per se has any direct significant actions on the sympathetic nervous system.

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